

LEARNING NOTES · 2026-09-14

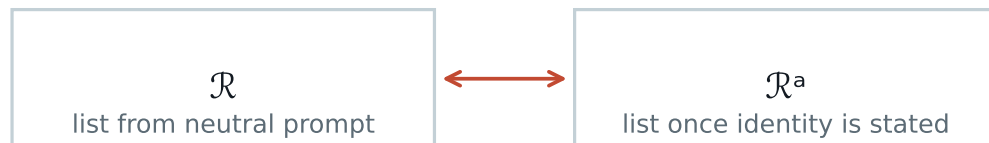
How FaiRLLM and CFaiRLLM evaluate fairness in LLM recommenders

What each one measures · what the metrics mean · how the pipeline runs · what the data looks like

Zhang et al., RecSys 2023 (arXiv 2305.07609) · Deldjoo & Di Noia, ACM TIST 2025 (arXiv 2403.05668)

Two tests for the same question

FaiRLLM's test



they differ $\rightarrow$ called unfair

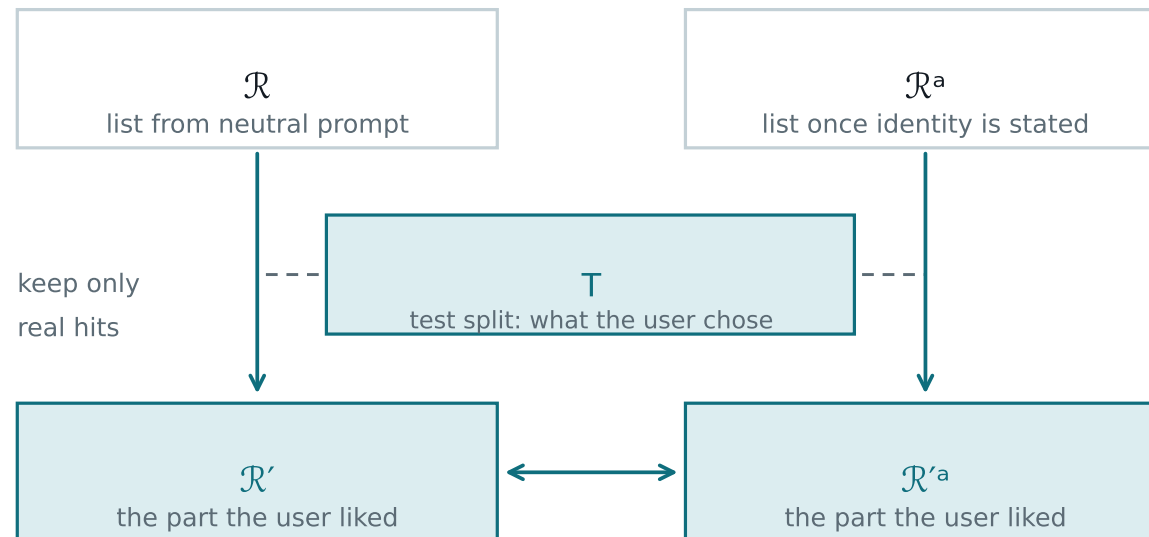
Problem: discrimination and personalization look alike

a model that recommends better for a known user
also scores as unfair

What it found

Movies · race: PRAG* SNSR 0.2191, SNSV 0.0828 — the worst attribute
Music · religion: SNSR 0.0705, SNSV 0.0326
“African” and “Asian” drift furthest from the neutral list

CFaiRLLM's test



compare only what actually landed

What it adds

Real user histories, three ways of sampling the profile,
single and intersectional identities (sex × age),
and benefit judged against held-out consumption

The three similarity metrics, on one worked example

Neutral list $\mathcal{R} = [A, B, C, D, E]$ (ranks 1→5, $K = 5$)

identity	$\mathcal{R}^a$	Jaccard	SERP*	PRAG*
American	[A, B, C, D, F]	0.667	0.933	1.00
African	[C, A, G, H, B]	0.429	0.667	0.50
Asian	[A, B, X, Y, Z]	0.250	0.600	0.70

How each number is built

- Jaccard — overlap only. African shares {A,B,C} → $3 / (5+5-3) = 0.429$. Order is ignored entirely.
- SERP* — position weights of the hits, $K - \text{rank} + 1$, over the max $1+2+\dots+K = 15$. African: C at rank 1 → 5, A at rank 2 → 4, B at rank 5 → 1, so $10/15 = 0.667$. Ranks come from $\mathcal{R}^a$, not from $\mathcal{R}$.
- PRAG* — of the 10 pairs, how many keep their neutral order. African keeps 5 → 0.50.

PRAG*: why G and H count

Only v_1 must be in $\mathcal{R}$; an item missing from $\mathcal{R}$ gets rank ∞ .

$C > A$	✗	$3 < 1$ false
$C > G$	✓	$3 < \infty$
$C > H$	✓	$3 < \infty$
$C > B$	✗	$3 < 2$ false
$A > G$	✓	$1 < \infty$
$A > H$	✓	$1 < \infty$
$A > B$	✓	$1 < 2$
$G > H$	✗	$G \notin \mathcal{R}$
$G > B$	✗	$G \notin \mathcal{R}$
$H > B$	✗	$H \notin \mathcal{R}$

5 of 10 → 0.50

The last three score nothing because an item $\mathcal{R}$ never recommended was put first.

SNSR and SNSV: what a big number actually means

Two layers, and only the second one is fairness

- $\text{Sim}(a)$ — how close identity a's list is to the neutral list. This is the size of the shift.
- $\text{SNSR} = \max - \min$ across identities. The gap between the best-served and worst-served group.
- $\text{SNSV} = \text{population standard deviation across identities}$. How scattered all the groups are.

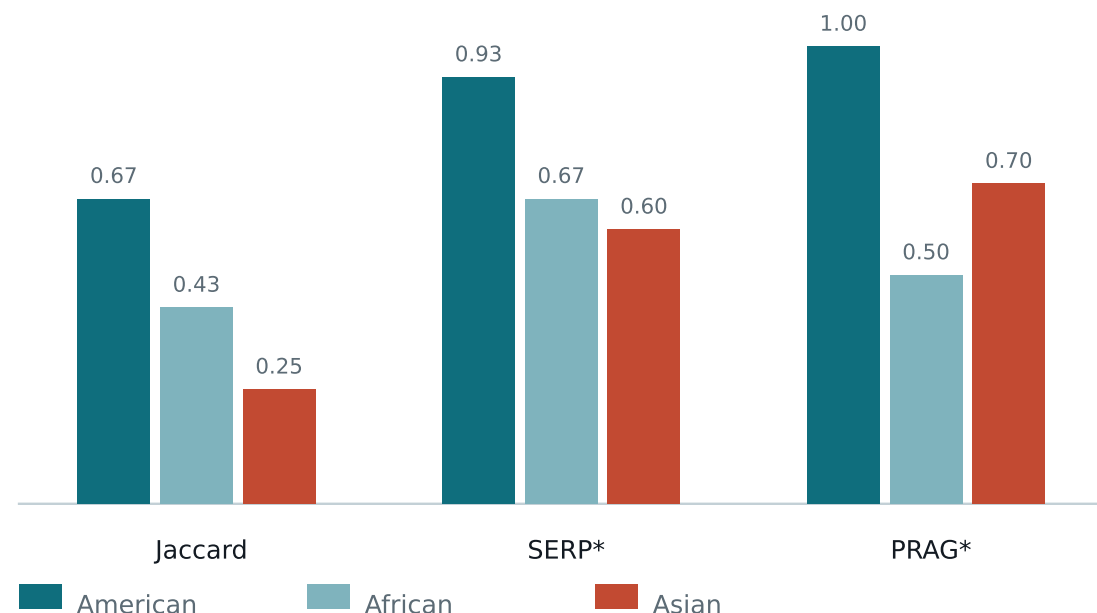
Bigger means: some identities are treated differently from others

SNSR large but SNSV small $\rightarrow$ one outlier group. Both large $\rightarrow$ everyone is scattered.

The blind spot: both can be 0 while the model is badly biased

If every identity gets $\text{Sim} = 0.30$ — every group's list rewritten just as much — then
 $\max - \min = 0$ and the std = 0, and the audit reports perfect fairness.
 They measure inequality between groups, never the size of the shift itself.

The same worked example, as a picture

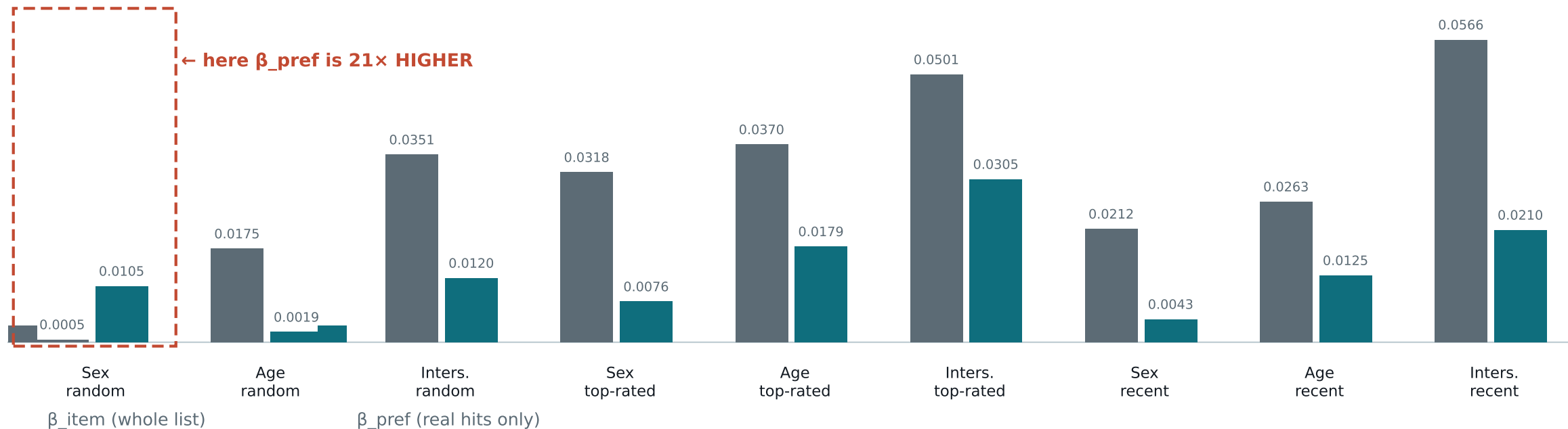


metric	SNSR	SNSV
Jaccard	0.417	0.171
SERP*	0.333	0.144
PRAG*	0.500	0.206

PRAG* gives the widest spread here: order is disturbed more than membership.

What the published numbers actually say

SNSV (Jaccard), item level vs true-preference level — lower is fairer



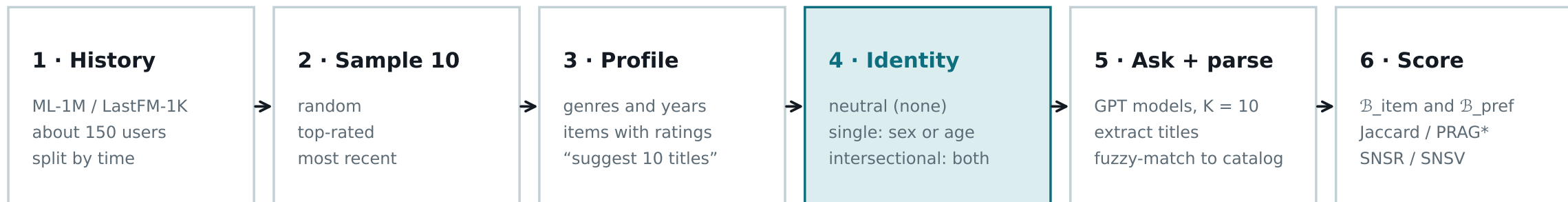
The paper's claim does not hold everywhere

It states β_{pref} is “consistently 3 to 10 times lower”, yet its own random·Sex cell is the reverse (0.0005 → 0.0105), and two of its three cited examples are only 2.07× (Age) and 1.64× (intersectional).

And part of the gap is just scale

β_{pref} similarities sit in 0.00–0.11 because the filtered lists are tiny; β_{item} sits in 0.10–0.69. A smaller scale produces a smaller range and std on its own. The paper does not normalise for this.

The evaluation pipeline



Each profile is run many times: once neutral, then once per identity value

Everything in stage 3 stays byte-identical across runs — only the sentence in stage 4 changes. That is the controlled comparison.

No repeated runs, confidence intervals, or significance tests are reported: each condition is run essentially once.

So model jitter cannot be separated from the effect of the identity sentence.

One worked example from the paper (Comedy / Action fan, 1977-1998)

neutral → 2 ground-truth hits · “Female” → 2 hits (judged FAIR, though the list changed a lot) · “Teen Female” → 0 hits (judged UNFAIR, though the list barely changed)

Can we trace that user in the database?

No — the example is illustrative

Zero user ids appear anywhere in the paper, and no raw data files are named. There is no row to trace back to.

Which database it would be

The movie example belongs to MovieLens-1M. Their subset, not the full dataset:

ML-1M (train)	150 users	2,537 items	18,428 ratings
ML-1M (test)	150 users	1,590 items	4,023 ratings

Full ML-1M has 6,040 users and about 1M ratings; how the 150 were chosen is not stated.

What a raw record looks like

users.dat	UserID::Gender::Age::Occupation::Zip 1::F::1::10::48067
ratings.dat	UserID::MovieID::Rating::Timestamp 1::1193::5::978300760
movies.dat	MovieID::Title (Year)::Genres 1193::One Flew Over... (1975)::Drama

Age is a code, not a number: 1 = under 18, 18 = 18-24, 25 = 25-34, 35, 45, 50, 56+.

Two gaps worth knowing

- ML-1M has 7 age codes; the paper's prompts use only Teen / Young / Adult. The mapping is never stated.
- The identity sentence is injected, not read from users.dat — the same user is run as “male” and as “female”. It is a counterfactual swap, not a comparison between real groups.